OsdagBridge

Open Source Software for Steel Girder Bridge Design

Design Report

Project Name	Composite Plate Girder Bridge
Project Location	Mumbai, Maharashtra
Author / Designer	Design Engineer
Reviewer	
Organization	OsdagBridge
Client Name and Organization	Public Works Department
Job Number	PG-30-7.5
Date	2026-08-19
Report Version	Rev 0

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Executive Summary

This section provides a concise summary of the bridge design, key inputs, governing loads, and final design outcomes.

Project Overview

Bridge Type	Composite Plate Girder Bridge
Design Standard	IRC 5, IRC 6, IRC 22, IRC 24, IS 800
Span	30 m
Carriageway Width	7.5 m
No. of Girders	4
Girder Spacing	2.5
Deck Thickness	220
Overall Design Status	Pass
Governing Check	Girder Moment
Overall Utilization Ratio (max)	0.82 (Girder)

[PLACEHOLDER: Figure 1 – Overall Bridge Plan]

Figure 1 – Overall Bridge Plan

[PLACEHOLDER: Figure 2 – Typical Cross-Section (with girder, deck, barriers, footpath)]

Figure 2 – Typical Cross-Section (with girder, deck, barriers, footpath)

[PLACEHOLDER: Figure 3 – 3D View of Bridge Superstructure]

Figure 3 – 3D View of Bridge Superstructure

Table 1 – Final Bridge Geometry (after optimization)

	G1	**G2**	**G3**	**G4**
Member ID	G1M1	G2M1	G3M1	G4M1
Section Designation	PG 1500x12 / 400x25 / 450x32	PG 1500x12 / 400x25 / 450x32	PG 1500x12 / 400x25 / 450x32	PG 1500x12 / 400x25 / 450x32
Governing Check	Girder Moment	Girder Moment	Girder Moment	Girder Moment
Utilization Ratio	0.82 (Girder)	0.82 (Girder)	0.82 (Girder)	0.82 (Girder)

Note: Utilization ratio (UR) = demand / capacity. A value < 1.0 *indicates a passing check.*

Key Design Outcomes Summary

Girder design pass
Cross bracing design pass
End Diaphragm design pass
Deck design pass

Design Assumptions and Limitations

- Additional inputs not provided by the user were assumed by software per IRC/IS code defaults or practical consideration.
- Grillage analysis was performed using OSPGrillage assuming simply supported I-girders.
- Substructure and foundation design are not included in this report.
- Splice connections and bearings are not designed in this version.

1 Project Information

This section records all project metadata as entered by the designer.

1.1 Project and Design Team Details

Project Name	Composite Plate Girder Bridge
Project Location	Mumbai, Maharashtra
Designer	Design Engineer
Reviewer	
Organization	OsdagBridge
Client	Public Works Department
Software Version	OsdagBridge

1.2 Applicable Codes and Standards

- Indian Roads Congress (IRC) 5: General Features of Design
- Indian Roads Congress (IRC) 6: Loads and Load Combinations
- Indian Roads Congress (IRC) 22: Composite Construction (Limit State Design)
- Indian Roads Congress (IRC) 24: Steel Road Bridges (Limit State Method)
- Indian Roads Congress (IRC) 112: Concrete Road Bridges (deck design)
- Indian Roads Congress Special Publication (IRC SP) 114: Seismic Design of Road Bridges
- Indian Standard (IS) 800: General Construction in Steel
- Indian Standard (IS) 2062: Hot Rolled Structural Steel Specification
- Indian Standard (IS) 6006: Steel Bearings

2 Input Parameters

This section documents all inputs provided to OsdagBridge. User-provided inputs are clearly distinguished from software-assumed defaults. Where the user did not supply a value, the software has applied the IRC/IS code default or an empirical guideline; these are annotated with an asterisk (*).

2.1 Basic Inputs (User-Defined)

Note: These inputs are mandatory and were provided by the user.

Table 2.1: Project Location

parameter	value
Project Location	Mumbai, Maharashtra
Latitude / Longitude	19.0760, 72.8777
Seismic Zone (IRC 6)	III
Basic Wind Speed (IRC 6)	39 m/s
Shade Temp. Max / Min (IRC 6)	45 °C / 5 °C

Table 2.2: Bridge Geometry

parameter	value
Type of Structure	Composite Plate Girder Bridge
Span (m)	30 m
Carriageway Width (m)	7.5 m
Include Median	No
Footpath	Both Sides
Skew Angle (degrees)	0° (IRC 24 Cl. 504.8 limit: ±15°)

Table 2.3: Material Selection

parameter	value
Girder Steel Grade (IS 2062)	E250 (Fe 410 W) A
Cross Bracing Steel Grade	E250 (Fe 410 W) A
End Diaphragm Steel Grade	E250 (Fe 410 W) A
Concrete Deck Grade (IRC 22)	M40

* *Software default value*

2.2 Additional Inputs

Where the user has modified additional inputs, those values are reported here. Where no modification was made, the software default is shown.

Table 2.4: Typical Section Details

parameter	value
Overall Bridge Width (m)	12.0
No. of Girders	4
Girder Spacing (m)	2.5 m
Deck Overhang Width (m)	1.0 m
Deck Thickness (mm)	220 mm
Footpath Width (m)	1.5 m (IRC 5 Cl. 104.3.6 min: 1.5 m)
No. of Traffic Lanes	2 (per IRC 5 Cl. 104.3.1)

Table 2.5: Components Details

parameter	value
Crash Barrier Type	IRC Concrete Barrier
Crash Barrier Load (kN/m)	5.0
Railing Type	Metal Beam
Railing Load (kN/m)	0.75
Wearing Course Material	Bituminous
Wearing Course Thickness (mm)	65 mm

Table 2.6: Girder General Information

Girder	Member ID	Design Mode	Girder Type	Girder Symmetry
G1	G1M1	Optimized	Welded I	Symmetric
G2	G2M1	Optimized	Welded I	Symmetric
G3	G3M1	Optimized	Welded I	Symmetric
G4	G4M1	Optimized	Welded I	Symmetric

Table 2.7: Girder Section Dimensions

Girder	Total Depth, D (mm)	Web, t_w (mm)	Top Flange (b_{tf}, t_{tf}) mm	Bottom Flange (b_{bf}, t_{bf}) mm
G1	1500 mm	12 mm	400 mm, 25 mm	450 mm, 32 mm
G2	1500 mm	12 mm	400 mm, 25 mm	450 mm, 32 mm
G3	1500 mm	12 mm	400 mm, 25 mm	450 mm, 32 mm
G4	1500 mm	12 mm	400 mm, 25 mm	450 mm, 32 mm

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Girder	**Torsional / Warping Restraint**	**Web Philosophy**	**Intermediate Stiffeners**	**Longitudinal Stiffeners**	**Bearing Stiffener**

Table 2.8: Girder Restraint and Stiffener Details

Girder	**Torsional / Warping Restraint**	**Web Philosophy**	**Intermediate Stiffeners**	**Longitudinal Stiffeners**	**Bearing Stiffener**
G1	Partial, Partial	Unstiffened	Yes; Spacing: 1500 mm; Thickness: 10 mm	No	No.: 2; Spacing: 300 mm; Thickness: 16 mm
G2	Partial, Partial	Unstiffened	Yes; Spacing: 1500 mm; Thickness: 10 mm	No	No.: 2; Spacing: 300 mm; Thickness: 16 mm
G3	Partial, Partial	Unstiffened	Yes; Spacing: 1500 mm; Thickness: 10 mm	No	No.: 2; Spacing: 300 mm; Thickness: 16 mm
G4	Partial, Partial	Unstiffened	Yes; Spacing: 1500 mm; Thickness: 10 mm	No	No.: 2; Spacing: 300 mm; Thickness: 16 mm

Table 2.9: Member Properties: Cross Bracing Details

Location	Member IDs	Type of Bracing	Bracing Section	Spacing (m)
G1-G2	B1M1	X-Bracing	75x75x8	5.0 m
G2-G3	B2M1	X-Bracing	75x75x8	5.0 m
G3-G4	B3M1	X-Bracing	75x75x8	5.0 m

Table 2.10: Member Properties: End Diaphragm Details

Location	Member IDs	Type of Bracing	Bracing Section
G1-G2	E1M1	Cross Bracing	75x75x8
G2-G3	E2M1	Cross Bracing	75x75x8
G3-G4	E3M1	Cross Bracing	75x75x8

Table 2.11: Shear Connector Details

parameter	value
Stud Diameter (mm)	22 mm
Stud Height (mm)	125 mm
Stud f_y (MPa)	350 MPa
Stud f_u (MPa)	450 MPa
No. of Studs per Section	2

Note: All values are per IRC 22 Table 1 unless user-modified.

Table 2.12: Partial Safety Factors

parameter	value
γ_{M0} (Yielding / Buckling)	1.10
γ_{M1} (Ultimate Stress)	1.25
γ_C (Concrete, Basic)	1.50
γ_s (Reinforcement)	1.15
γ_v (Shear Connectors)	1.25
γ_{fft} (Fatigue Load)	1.0
γ_{Mft} (Fatigue Strength)	1.0

3 Loads and Load Combinations

This section summarizes all loads applied to the bridge and the load combinations considered for analysis and design.

Table 3.1: Dead Load – Self Weight

parameter	value
Steel Self-Weight Applied	78.5 kN/m^3
Concrete Deck Weight	25 kN/m^3
Self-Weight Factor	1.05

Table 3.2: Dead Load for Surfacing (DW)

parameter	value
Wearing Course Load	Bituminous x 65
Additional SIDL (Crash Barrier)	5.0 kN/m per barrier
Railing Load	0.75 kN/m*

Table 3.3: Vehicle Live Loads (LL)

Vehicle	Total Load (kN)	Impact Factor	Braking Load		
			Considered?	Value	Eccentricity
Class A	543.47	1.207	Yes	108.69	1.20
Class 70R	981.00	1.207	Yes	108.69	1.20
Class SV	3776.85	1.207	No	—	—

Note: Vehicle rows follow Additional Inputs → Loading tab selections. Total load and impact factor (1 + IM) *per IRC 6 Cl. 204 / Cl. 208. Braking force is the carriageway value per Cl. 211.2 (shown on each vehicle for which braking is considered). Eccentricity is measured from the deck top per Cl. 211.3.*

* *Software default value*

Table 3.4: Centrifugal Force (IRC 6 Cl. 212)

Vehicle	Considered?	Value (kN)	Radius (m)	Speed (km/h)
Class A	Yes	109.55	250.00	80
Class 70R	Yes	197.74	250.00	80
Class SV	Yes	761.32	250.00	80

Note: Centrifugal force $F = Wv^2/(127R)$ per IRC 6 Cl. 212 is applied only when a horizontal curve radius is provided. Straight bridges are marked Not considered.

Table 3.5: Footpath Live Loads

Parameter	Value	Unit	Reference / Note
Footpath configuration	Both Sides	—	—
Footpath live load applicable?	Yes	—	—
Pressure mode	As per IRC 6 Cl. 206.1	—	Default footway pressure
Footway live load intensity	4.905	kN/m^2	IRC 6 Cl. 206.1

Table 3.6: Wind Load (WL) — per IRC 6

parameter	value
Basic Wind Speed, Vb	39 m/s [from Project Location]
Terrain Type	Category II
Average Exposed Height, H (m)	2.5 m
Hourly Mean Wind Speed, Vz	
Hourly Wind Pressure, Pz	
Transverse Wind Force	
Longitudinal Wind Force	
Vertical Wind Force	

Table 3.7: Earthquake Load (EL) — per IRC 6

parameter	value
Seismic Zone	III [from Project Location]
Zone Factor, Z	
Importance Factor, I	1.0
Type of Soil	Medium
Sa/g	
Horizontal Seismic Coefficient, Ah	
Vertical Seismic Coefficient, Av	
Horizontal Seismic Force (longitudinal)	kN
Horizontal Seismic Force (transverse)	kN

Table 3.8: Temperature Load (TL) — per IRC 6

parameter	value
Maximum Shade Temperature	45 °C
Minimum Shade Temperature	5 °C
Effective Bridge Temp. Range	to °C
Temperature Rise / Fall for Design	+ °C / − °C

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Table 3.9: Load Combinations (continued)

Combination ID	Load Cases

Table 3.9: Load Combinations

Combination ID	Load Cases
ULS-01	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(1.50) + WL(0.90) + TL(0.90)
ULS-02	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(1.15) + WL(1.50) + TL(0.90)
ULS-03	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(1.15) + WL(0.90) + TL(1.50)
ULS-04	DL(1.00/1.00) + SIDL(1.00/1.00) + LL(0.75) + TL(0.50) + VC(1.00)
ULS-05	DL(1.00/1.00) + SIDL(1.00/1.00) + LL(0.75) + TL(0.50) + BI(1.00)
ULS-06	DL(1.00/1.00) + SIDL(1.00/1.00) + LL(0.75) + TL(0.50) + FB(1.00)
ULS-07	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(0.20) + TL(0.50) + EQ(1.50)
ULS-08	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(0.20) + TL(0.50) + EQ(0.75)
SLS-01	DL(1.00) + SIDL(1.20/1.00) + LL(1.00) + WL(0.60) + TL(0.60)
SLS-02	DL(1.00) + SIDL(1.20/1.00) + LL(0.75) + WL(1.00) + TL(0.60)
SLS-03	DL(1.00) + SIDL(1.20/1.00) + LL(0.75) + WL(0.60) + TL(1.00)
SLS-04	DL(1.00) + SIDL(1.20/1.00) + LL(0.75) + WL(0.50) + TL(0.50)
SLS-05	DL(1.00) + SIDL(1.20/1.00) + LL(0.20) + WL(0.60) + TL(0.50)
SLS-06	DL(1.00) + SIDL(1.20/1.00) + LL(0.20) + WL(0.50) + TL(0.60)
SLS-07	DL(1.00) + SIDL(1.20/1.00) + LL(0.00) + WL(0.00) + TL(0.50)

Note: All IRC 6 load combinations are auto-generated by OsdagBridge. User-defined custom combinations, if any, are appended.

4 Analysis Results

A grillage model was used for structural analysis. The deck is idealized as a grid of elastic beam elements — longitudinal members represent the composite steel girders with effective slab, and transverse members represent the slab or cross frames. This section summarizes the critical output from that analysis.

Table 4.1: Summary of Maximum Demands

Load Case/ Comb.	Bending Moment			Shear Force			Reaction at Supports	
	Max (kNm)	Girder	Loc. (m)	Max (kN)	Girder	Loc. (m)	Left	Right
DL	820.000	G1	15.000	140.000	G1	0.500	320.000	320.000
Basic ULS: 1.35DL + 1.5LL	2450.000	G1	15.000	420.000	G1	0.500	980.000	980.000
ACCIDENTAL 1: 1.0DL + 0.5LL	1580.000	G2	14.500	290.000	G2	0.500	610.000	605.000

Table 4.2: Reactions at Supports

Load Case	Left Support (kN)	Right Support (kN)
DL	320.000	320.000
Basic ULS: 1.35DL + 1.5LL	980.000	980.000
ACCIDENTAL 1: 1.0DL + 0.5LL	610.000	605.000

Table 4.3: Deflection Summary (Live Load & Total Load)

parameter	value
Deflection due to Live Load, δ_{LL}	28.500 mm
Allowable Live Load Deflection (Δ_{allow})	L/800 = 37.5 mm
Live Load Deflection Check Status	PASS
Deflection due to Total Load, δ_{total}	42.000 mm
Allowable Total Deflection (Δ_{allow})	L/600 = 50.0 mm
Total Load Deflection Check Status	PASS

[PLACEHOLDER: Bending Moment Envelope (Envelope ULS): Max/min BM along span. X-axis: distance from left support (m). Y-axis: Bending Moment (kN-m).]

[PLACEHOL X-axis: distan

[PLACEHOLDER: Vertical Deflection D_y (1.0 LL): Maximum deflection along span. Load Case: 1.0 LL, Combination: D_y. Nodes shown. Isometric view.]

5 Design Checks

This section presents all structural design checks performed by OsdagBridge. For each member, the demand from the governing load combination, the code-based capacity, and the utilization ratio are tabulated. All checks reference IS 800:2007 and IRC 22:2014 unless stated otherwise.

5.1 Plate Girder Design

Table 5.1: Girder Section Properties (Final Optimized / User-selected)

Girder	Property	Value
G1	Depth, D (mm)	1500
	Top Flange Width, b_f (mm)	400
	Bottom Flange Width, b_f (mm)	450
	Top Flange Thickness, t_f (mm)	25
	Bottom Flange Thickness, t_f (mm)	32
	Web Thickness, t_w (mm)	12
	Gross Area, A (cm^2)	285
	Moment of Inertia, I_z (cm^4)	1850000.0
	Elastic Section Modulus, Z_{ez} (cm^3)	24500
	Plastic Section Modulus, Z_{pz} (cm^3)	28000
	Effective Slab Width, b_{eff} (mm)	2500
	Transformed Composite I_z (cm^4)	3100000.0
	Depth to Plastic Neutral Axis (mm)	185
G2	Depth, D (mm)	1500
	Top Flange Width, b_f (mm)	400
	Bottom Flange Width, b_f (mm)	450
	Top Flange Thickness, t_f (mm)	25

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Table 5.1: Girder Section Properties (Final Optimized / User-selected) (continued)

Girder	Property	Value
	Bottom Flange Thickness, t_f (mm)	32
	Web Thickness, t_w (mm)	12
	Gross Area, A (cm^2)	285
	Moment of Inertia, I_z (cm^4)	1850000.0
	Elastic Section Modulus, Z_{ez} (cm^3)	24500
	Plastic Section Modulus, Z_{pz} (cm^3)	28000
	Effective Slab Width, b_{eff} (mm)	2500
	Transformed Composite I_z (cm^4)	3100000.0
	Depth to Plastic Neutral Axis (mm)	185
G3	Depth, D (mm)	1500
	Top Flange Width, b_f (mm)	400
	Bottom Flange Width, b_f (mm)	450
	Top Flange Thickness, t_f (mm)	25
	Bottom Flange Thickness, t_f (mm)	32
	Web Thickness, t_w (mm)	12
	Gross Area, A (cm^2)	285
	Moment of Inertia, I_z (cm^4)	1850000.0
	Elastic Section Modulus, Z_{ez} (cm^3)	24500
	Plastic Section Modulus, Z_{pz} (cm^3)	28000
	Effective Slab Width, b_{eff} (mm)	2500
	Transformed Composite I_z (cm^4)	3100000.0
	Depth to Plastic Neutral Axis (mm)	185
G4	Depth, D (mm)	1500

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Table 5.1: Girder Section Properties (Final Optimized / User-selected) (continued)

Girder	Property	Value
	Top Flange Width, b_f (mm)	400
	Bottom Flange Width, b_f (mm)	450
	Top Flange Thickness, t_f (mm)	25
	Bottom Flange Thickness, t_f (mm)	32
	Web Thickness, t_w (mm)	12
	Gross Area, A (cm^2)	285
	Moment of Inertia, I_z (cm^4)	1850000.0
	Elastic Section Modulus, Z_{ez} (cm^3)	24500
	Plastic Section Modulus, Z_{pz} (cm^3)	28000
	Effective Slab Width, b_{eff} (mm)	2500
	Transformed Composite I_z (cm^4)	3100000.0
	Depth to Plastic Neutral Axis (mm)	185

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Table 5.2: Girder Section Classification (continued)

	Element	Slenderness Ratio	Class Limit	Classification

Table 5.2: Girder Section Classification

	Element	Slenderness Ratio	Class Limit	Classification
G1	Top Flange	8.0	9.4	Plastic
	Web	120.0	126.0	Compact
	Overall Section	—	—	
G2	Top Flange	8.0	9.4	Plastic
	Web	120.0	126.0	Compact
	Overall Section	—	—	
G3	Top Flange	8.0	9.4	Plastic
	Web	120.0	126.0	Compact
	Overall Section	—	—	
G4	Top Flange	8.0	9.4	Plastic
	Web	120.0	126.0	Compact
	Overall Section	—	—	

Note: IS 800:2007 Table 2

Table 5.3: Moment Capacity Check

	Parameter	Formula	Value	Status
G1	Applied Moment, M_u	Governing LC (ULS)	2450 kN-m	—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1	3000 kN-m	—
	Utilization Ratio, M_u/M_d	M_u/M_d	0.82	PASS

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Table 5.3: Moment Capacity Check (continued)

	Parameter	Formula	Value	Status
G2	Applied Moment, M_u	Governing LC (ULS)	2450 kN-m	—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1	3000 kN-m	—
	Utilization Ratio, M_u/M_d	M_u/M_d	0.82	PASS
G3	Applied Moment, M_u	Governing LC (ULS)	2450 kN-m	—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1	3000 kN-m	—
	Utilization Ratio, M_u/M_d	M_u/M_d	0.82	PASS
G4	Applied Moment, M_u	Governing LC (ULS)	2450 kN-m	—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1	3000 kN-m	—
	Utilization Ratio, M_u/M_d	M_u/M_d	0.82	PASS

Note: IRC 22 Cl. 603.3.1, IS 800 Cl. 8.2.1

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Table 5.4: **Shear Capacity Check** (continued)

	Parameter	Formula	Value	Status

Table 5.4: Shear Capacity Check

	Parameter	Formula	Value	Status
G1	Applied Shear, V_u	Governing LC (ULS)	420 kN	—
	Shear Area, A_v	$d_w \times t_w$	17280 mm^2	—
	Panel Aspect Ratio, c/d	—	1.0	—
	Shear Buckling Coefficient, k_v	IS 800 Cl. 8.4.2.2	5.34	—
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\,\tau_{cr})}$	0.85	—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2	95.0 MPa	—
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$	690 kN	—
	Utilization Ratio, V_u/V_d	V_u/V_d	0.61	PASS
G2	Applied Shear, V_u	Governing LC (ULS)	420 kN	—
	Shear Area, A_v	$d_w \times t_w$	17280 mm^2	—
	Panel Aspect Ratio, c/d	—	1.0	—
	Shear Buckling Coefficient, k_v	IS 800 Cl. 8.4.2.2	5.34	—
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\,\tau_{cr})}$	0.85	—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2	95.0 MPa	—
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$	690 kN	—

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Table 5.4: Shear Capacity Check (continued)

	Parameter	**Formula**	**Value**	**Status**
	Utilization Ratio, V_u/V_d	V_u/V_d	0.61	PASS
G3	Applied Shear, V_u	Governing LC (ULS)	420 kN	—
	Shear Area, A_v	$d_w \times t_w$	17280 mm^2	—
	Panel Aspect Ratio, c/d	—	1.0	—
	Shear Buckling Coefficient, k_v	IS 800 Cl. 8.4.2.2	5.34	—
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\,\tau_{cr})}$	0.85	—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2	95.0 MPa	—
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$	690 kN	—
	Utilization Ratio, V_u/V_d	V_u/V_d	0.61	PASS
G4	Applied Shear, V_u	Governing LC (ULS)	420 kN	—
	Shear Area, A_v	$d_w \times t_w$	17280 mm^2	—
	Panel Aspect Ratio, c/d	—	1.0	—
	Shear Buckling Coefficient, k_v	IS 800 Cl. 8.4.2.2	5.34	—
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\,\tau_{cr})}$	0.85	—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2	95.0 MPa	—
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$	690 kN	—
	Utilization Ratio, V_u/V_d	V_u/V_d	0.61	PASS

Note: IS 800 Cl. 8.4, IRC 22 Cl. 603.3.3.2

Table 5.5: Interaction Checks (M-V and M-N)

	Check	Condition	Value	Status
G1	High Shear Condition?	$V_u > 0.6\,V_d$		—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3		—
	Interaction Check: $M_u \leq M_{dv}$	—	0.55	PASS
	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	N/A	N/A	N/A
G2	High Shear Condition?	$V_u > 0.6\,V_d$		—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3		—
	Interaction Check: $M_u \leq M_{dv}$	—	0.55	PASS
	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	N/A	N/A	N/A
G3	High Shear Condition?	$V_u > 0.6\,V_d$		—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3		—
	Interaction Check: $M_u \leq M_{dv}$	—	0.55	PASS
	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	N/A	N/A	N/A
G4	High Shear Condition?	$V_u > 0.6\,V_d$		—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3		—
	Interaction Check: $M_u \leq M_{dv}$	—	0.55	PASS

Continued on next page

Table 5.5: Interaction Checks (M-V and M-N) (continued)

	Check	**Condition**	**Value**	**Status**
	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	N/A	N/A	N/A

Note: IRC 22 Cl. 603.3.3.3

Continued on next page

Table 5.6: Lateral Torsional Buckling Check – Construction Stage (continued)

Table 5.6: Lateral Torsional Buckling Check – Construction Stage

	Parameter	**Formula**	**Value**	**Status**
G1	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1	5200 kN-m	—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$	0.76	—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2	0.78	—
	LTB Resistance, M_b	$\chi_{LT}\, M_p/\gamma_{m0}$	2340 kN-m	—
	$M_u \leq M_b$	M_u/M_b	0.54	PASS
G2	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1	5200 kN-m	—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$	0.76	—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2	0.78	—
	LTB Resistance, M_b	$\chi_{LT}\, M_p/\gamma_{m0}$	2340 kN-m	—
	$M_u \leq M_b$	M_u/M_b	0.54	PASS
G3	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1	5200 kN-m	—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$	0.76	—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2	0.78	—
	LTB Resistance, M_b	$\chi_{LT}\, M_p/\gamma_{m0}$	2340 kN-m	—
	$M_u \leq M_b$	M_u/M_b	0.54	PASS
G4	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1	5200 kN-m	—

Continued on next page

Table 5.6: Lateral Torsional Buckling Check – Construction Stage (continued)

	Parameter	**Formula**	**Value**	**Status**
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$	0.76	—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2	0.78	—
	LTB Resistance, M_b	$\chi_{LT}\, M_p/\gamma_{m0}$	2340 kN-m	—
	$M_u \leq M_b$	M_u/M_b	0.54	PASS

Note: IRC 22 Cl. 603.3.3.1, IS 800 Cl. 8.2.2

Continued on next page

Table 5.7: Stiffener Design Summary (continued)

Table 5.7: Stiffener Design Summary

G1	Shear Buckling Design Method	
	Intermediate Stiffener Thickness (mm)	
	Intermediate Stiffener Spacing (mm)	
	End Panel Stiffener Thickness (mm)	
	No. of End Panel Stiffeners	
	Longitudinal Stiffeners	
G2	Shear Buckling Design Method	
	Intermediate Stiffener Thickness (mm)	
	Intermediate Stiffener Spacing (mm)	
	End Panel Stiffener Thickness (mm)	
	No. of End Panel Stiffeners	
	Longitudinal Stiffeners	
G3	Shear Buckling Design Method	
	Intermediate Stiffener Thickness (mm)	
	Intermediate Stiffener Spacing (mm)	
	End Panel Stiffener Thickness (mm)	
	No. of End Panel Stiffeners	
	Longitudinal Stiffeners	
	Shear Buckling Design Method	

Continued on next page

G4

Table 5.7: Stiffener Design Summary (continued)

	Intermediate Stiffener Thickness (mm)	
	Intermediate Stiffener Spacing (mm)	
	End Panel Stiffener Thickness (mm)	
	No. of End Panel Stiffeners	
	Longitudinal Stiffeners	

Continued on next page

Table 5.8: End Panel Stiffener Checks (continued)

	Check	**Required**	**Provided**	**Status**

Table 5.8: End Panel Stiffener Checks

	Check	**Required**	**Provided**	**Status**
G1	Web Buckling Resistance	N/A	N/A	N/A
	Local Crushing Resistance	N/A	N/A	N/A
	Bearing Capacity	N/A	N/A	N/A
	Column Buckling Resistance	N/A	N/A	N/A
G2	Web Buckling Resistance	N/A	N/A	N/A
	Local Crushing Resistance	N/A	N/A	N/A
	Bearing Capacity	N/A	N/A	N/A
	Column Buckling Resistance	N/A	N/A	N/A
G3	Web Buckling Resistance	N/A	N/A	N/A
	Local Crushing Resistance	N/A	N/A	N/A
	Bearing Capacity	N/A	N/A	N/A
	Column Buckling Resistance	N/A	N/A	N/A
G4	Web Buckling Resistance	N/A	N/A	N/A
	Local Crushing Resistance	N/A	N/A	N/A
	Bearing Capacity	N/A	N/A	N/A
	Column Buckling Resistance	N/A	N/A	N/A

Note: IS 800 Cl. 8.4.2.2

Table 5.9: Serviceability – Deflection Checks

	Check	Allowable	Actual	Status
G1	Live Load Deflection, δ_{LL} (mm)	L/800 = 37.5 mm	28.500	PASS
	Total Load Deflection, δ_{total} (mm)	L/600 = 50.0 mm	42.000	PASS
G2	Live Load Deflection, δ_{LL} (mm)	L/800 = 37.5 mm	27.200	PASS
	Total Load Deflection, δ_{total} (mm)	L/600 = 50.0 mm	40.500	PASS
G3	Live Load Deflection, δ_{LL} (mm)	L/800 = 37.5 mm		—
	Total Load Deflection, δ_{total} (mm)	L/600 = 50.0 mm		—
G4	Live Load Deflection, δ_{LL} (mm)	L/800 = 37.5 mm		—
	Total Load Deflection, δ_{total} (mm)	L/600 = 50.0 mm		—

Note: IRC 22 Cl. 604.3.2

Continued on next page

Table 5.10: Serviceability – Maximum Stress Limitation (continued)

	Element	**Allowable Stress**	**Actual Stress**	**Status**

Table 5.10: Serviceability – Maximum Stress Limitation

	Element	**Allowable Stress**	**Actual Stress**	**Status**
G1	Structural Steel ($0.9\ f_y$)			—
G2	Structural Steel ($0.9\ f_y$)			—
G3	Structural Steel ($0.9\ f_y$)			—
G4	Structural Steel ($0.9\ f_y$)			—

Table 5.11: Serviceability – Fatigue Assessment

	Stress Range, $\Delta\sigma$ (MPa)	**Fatigue Limit, f_{fd} (MPa)**	**Utilization Ratio**	**Status**
G1				—
G2				—
G3				—
G4				—

Note: IRC 22 Cl. 605 — governing of normal and shear fatigue (worst by DCR). Capacity reduction factor μ_r applied where plate thickness ¿ 25 mm.

Table 5.12: Girder Design Summary (DCR / Utilization Ratio)

Girder	**Controlling LC / Combination**	**Controlling Check**	**Demand**	**Capacity**	**UR**	**Status**
G1	Basic ULS: 1.35DL + 1.5LL	Girder Moment	2450.00 kNm	3000.00 kNm	0.820	—
G2	Basic ULS: 1.35DL + 1.5LL	Girder Moment	2370.00 kNm	3000.00 kNm	0.790	—
G3						

Continued on next page

Table 5.12: Girder Design Summary (DCR / Utilization Ratio) (continued)

Girder	Controlling LC / Combination	Controlling Check	Demand	Capacity	UR	Status
G4						

Note: UR = Demand / Capacity. A value ≤ 1.0 indicates a passing check. The controlling check is the criterion with the highest UR for each girder, with the real load case/combination that drives it.

Table 5.13: Shear Connector Capacity

Parameter	Formula	Value	Reference
Design Resistance, Q_u	$Q_u = \min(Q_{u,s},\, Q_{u,c})$ $Q_{u,s} = \dfrac{0.8\, f_u\, (\pi d^2/4)}{\gamma_v}$ $Q_{u,c} = \dfrac{0.29\, \alpha\, d^2 \sqrt{f_{ck}\, E_{cm}}}{\gamma_v}$		IRC 22 Cl. 606.3.1 (Eq. 6.1)
Fatigue Shear Resistance, Q_r	IRC 22 Table 8 (ϕd, N_{sc})		IRC 22 Cl. 606.3.2 (Table 8)

Table 5.14: Shear Connector Spacing

Criterion	Governing Spacing	Actual Spacing Provided	Status
ULS Shear (SL1)			—
Full Composite (SL2)			—
SLS Fatigue (SR)			—
Max Spacing Limit (IRC 22)			—

Note: IRC 22 Cl. 606.4, 606.9. Governing spacing $= \min(S_{L1}, S_{L2}, S_R)$.

Table 5.15: Transverse Shear and Detailing Checks

Check	Value	Status
Longitudinal Shear per unit length, V_L		—
Transverse Shear Capacity of Slab, V_{Rd}		—
Transverse Shear Check	$V_L/V_{Rd} =$ —	—
Min. Transverse Reinforcement, $A_{st,min}$	Required , Provided 0.00 cm^2/m	—
Stud Diameter $\leq 2\,t_f$	$d = \leq 2t_f =$	—
Stud Edge Distance	Provided (req. $\geq$)	—

Note: IRC 22 Cl. 606.6, 606.10.

5.2 Deck Slab Design

The reinforced concrete deck slab is designed per IRC 112:2011 (flexure, shear, crack width) and IRC 22:2014 (composite construction). Wheel loads are distributed using Pigeaud's method. The deck is checked for flexure in the transverse and longitudinal directions, punching shear, one-way (beam) shear, crack width, and reinforcement detailing.

Table 5.16: Deck Slab — Loading and Geometry

Effective Span of Deck Slab, l_{eff}	2500 mm (girder spacing, c/c)
Deck Thickness, t_s	220 mm
Clear Cover (IRC 112 Cl. 15.2)	Top / Bottom mm
Concrete Grade (IRC 112 Cl. 6.4)	M40 ($f_{ck} =$ MPa, $f_{ctm} =$ MPa)
Reinforcement Grade (IRC 112 Cl. 6.2)	($f_y = <—>$ MPa)
Dead Load per Unit Area, w_{DL}	5.50 kN/m^2 (slab self-weight)

Continued on next page

Table 5.16: Deck Slab — Loading and Geometry (continued)

IRC 6 Wheel Load (Class A / 70R)	<—> kN
Tyre Contact Width (IRC 6 Annex A)	<—> mm (transverse)
Impact Factor (IRC 6 Cl. 208.2)	<—>
Governing Live Load Case	<—>

Table 5.17: Deck Slab — Flexure Check: Interior Panel (Pigeaud's Method)

Location	**Parameter**	**Formula / Reference**	**Value**	**Status**
At Midspan (Sagging)	Transverse BM (DL), $M_{T,DL}$	$w_{DL}\, l_{eff}^2/10$	8.20 kN-m/m	—
	Transverse BM (LL), $M_{T,LL}$	Effective width (IRC 112 B3.1)	<—> kN-m/m	—
	Total Design BM, $M_{u,sag}$	1.35 DL + 1.50 LL	42.50 kN-m/m	—
	Effective depth, d	$t_s - c_{nom} - \phi/2$	<—> mm	—
	Moment Capacity, M_{Rd}	IRC 112 Cl. 12.2	63.00 kN-m/m	PASS
At Support (Hogging)	Total Design BM, $M_{u,hog}$	1.35 DL + 1.50 LL (at support)	38.00 kN-m/m	—
	Required Top Steel, $A_{st,top}$	$M_u/(0.87\, f_y\, d)$	<—> mm²/m	—
	Moment Capacity, M_{Rd}	IRC 112 Cl. 12.2	58.00 kN-m/m	PASS

Note: IRC 112 Cl. 12.2. Distribution (longitudinal) reinforcement designed for 20% of main steel moment (IRC 21 Cl. 305.18).

Table 5.18: Deck Slab — Cantilever Overhang Flexure Check

Parameter	**Formula**	**Value**	**Status**
Overhang Length, l_{oh}	—	1.0 m	—
Crash Barrier Load Moment	IRC 6 Cl. 206.4	<—> kN-m/m	—
Dead Load Moment	$w_{DL}\, l_{oh}^2/2$ + railing	<—> kN-m/m	—
Live Load Moment (eccentric wheel)	Wheel load × arm	<—> kN-m/m	—
Total Hogging Moment, $M_{u,oh}$	1.35 DL + 1.50 (LL + CB)	18.50 kN-m/m	—
Moment Capacity (top steel), $M_{Rd,oh}$	IRC 112 Cl. 12.2	29.00 kN-m/m	PASS

Note: IRC 6 Cl. 206.4 crash barrier loads applied at kerb face; IRC 112 Cl. 12.2 flexure.

Table 5.19: Deck Slab — Punching Shear Check (IRC 112 Cl. 10.4.6)

Parameter	**Formula / Reference**	**Value**	**Status**
Design Wheel Load (ULS), V_{Ed}	$\gamma_Q\,(1 + IF)\,P_w$	<—> kN	—
Tyre Contact Area	$a \times b$ (IRC 6 Annex A)	<—> × <—> mm	—
Loaded Area at mid-depth, b_0	$c_1 \times c_2$ (incl. WC dispersion)	<—> × <—> mm	—
Control Perimeter, u_1	$2(c_1 + c_2) + 4\pi d$	<—> mm	—
Punching Shear Stress, v_{Ed}	$V_{Ed}/(u_1\, d)$	0.850 MPa	—
Punching Resistance, $v_{Rd,c}$	IRC 112 Eq. 10.1	1.250 MPa	—
Punching Shear Check	$v_{Ed} \leq v_{Rd,c}$	0.68	FAIL

Note: Punching shear reinforcement not typically required for deck slabs with $d \geq 200$ mm and adequate longitudinal reinforcement.

Table 5.20: Crack Width Check (Deck Slab)

Parameter	**Value / Reference**
Min. Reinforcement for Crack Control, $A_{s,min}$	<—> mm²/m [IRC 112 Cl. 16.5.1]
Provided Reinforcement (bottom)	ϕ<—> @ <—> mm c/c (<—> mm²/m)
Max. Permissible Crack Width	0.30 mm
Calculated Crack Width, w_k (governing)	0.1800 mm
Crack Width Check	PASS

Table 5.21: One-Way (Beam) Shear Check (Deck Slab)

Parameter	**Formula / Reference**	**Value**	**Status**
Design Shear per unit width, V_{Ed}	$\gamma_{DL} V_{DL} + \gamma_{LL}(1{+}IF) V_{LL}$	95.00 kN/m	—
Effective depth, d	$t_s - c_{nom} - \phi/2$	<—> mm	—
Size factor, k	$1 + \sqrt{200/d} \leq 2.0$	<—>	—
Long. reinforcement ratio, ρ_l	$A_{sl}/(b_w\, d) \leq 0.02$	<—>	—
Shear resistance (no stirrups), $V_{Rd,c}$	$v_{Rd,c}\, b_w\, d$ (Cl. 10.3.2)	140.00 kN/m	—
One-Way Shear Check	$V_{Ed} \leq V_{Rd,c}$	0.68	FAIL

Note: IRC 112 Cl. 10.3.2. Shear reinforcement not provided in deck slabs; capacity relies on concrete and main reinforcement.

Continued on next page

Table 5.22: Reinforcement Detailing Summary (Deck Slab) (continued)

Parameter	**Required / Limit**	**Provided**	**Status**

Table 5.22: Reinforcement Detailing Summary (Deck Slab)

Parameter	**Required / Limit**	**Provided**	**Status**
Main Reinforcement — Bottom (Transverse)			
Required Area, $A_{st,req}$ (mm²/m)	<—> mm²/m	<—> mm²/m	PASS
Bar Diameter × Spacing	$\phi \geq 10$ mm (IRC 112)	ϕ<—> @ <—> mm c/c	—
Min. Reinforcement $A_{s,min}$ (IRC 112 Cl. 16.3.1)	<—> mm²/m	<—> mm²/m	PASS
Max. Bar Spacing (IRC 112 Cl. 16.3.2)	<—> mm	<—> mm	FAIL
Distribution Reinforcement — Longitudinal			
Required Area, $A_{st,dist}$ (mm²/m)	$\geq$ 20% of main steel	<—> mm²/m	PASS
Top Reinforcement (Support / Cantilever Overhang)			
Required Area, $A_{st,top}$ (mm²/m)	<—> mm²/m	<—> mm²/m	PASS
Cover and Detailing			
Clear Cover (IRC 112 Cl. 15.2)	$\geq$ <—> mm (Table 14.2)	Top / Bottom mm	FAIL

Note: IRC 112 Cl. 16.3, IS 456 Cl. 26.5. All reinforcement provisions satisfy strength and detailing requirements.

5.3 Cross Bracing Design

Cross bracing between adjacent plate girders provides lateral stability during construction, resists transverse loads (wind, seismic, braking) in service, and prevents lateral torsional buckling of the girders. Members are designed per IS 800:2007 Cl. 7 (compression) and Cl. 6 (tension). Forces are derived from the grillage model under the governing load combination (DL + LL + WL).

Table 5.23: Cross Bracing — Connection and Section Properties

Panel	Member	Connection	Section	A_g (mm²)	r_{min} (mm)
G1-G2	Diagonal	Welded	75x75x8		
G1-G2	Top / Bottom chord	Welded	65x65x6		
G2-G3	Diagonal	Welded	75x75x8		
G2-G3	Top / Bottom chord	Welded	65x65x6		

Note: A_g = gross cross-sectional area; r_{min} = minimum radius of gyration.

Table 5.24: Cross Bracing — Slenderness Ratio Check (IS 800 Cl. 3.8)

Panel	Member	Nature	Eff. Length KL (mm)	KL/r	Limit / Status
G1-G2	Diagonal	C	2920	142.0	250 — PASS
	Top chord	C	2500	118.0	250 — PASS
	Bottom chord	T	2500	118.0	400 — PASS
G2-G3	Diagonal	C	2920	142.0	250 — PASS
	Top chord	C	2500	118.0	250 — PASS
	Bottom chord	T	2500	118.0	400 — PASS

Note: IS 800 Cl. 3.8 — Limit = 250 *for compression members,* 400 *for tension members.* $K = 1.0$ *for members with both ends pinned.*

Table 5.25: Cross Bracing Design — Capacity Summary

Panel	Member	Section	Governing LC	Demand (kN)	Capacity (kN)	UR	Status
G1-G2	Diagonal	75x75x8	Basic ULS: 1.35DL + 1.5LL	185.000	340.000	0.540	PASS
	Chord	65x65x6	Basic ULS: 1.35DL + 1.5LL	95.000	210.000	0.450	PASS
G2-G3	Diagonal	75x75x8	Basic ULS: 1.35DL + 1.5LL	172.000	340.000	0.510	PASS
	Chord	65x65x6	Basic ULS: 1.35DL + 1.5LL	88.000	210.000	0.420	PASS

Note: Designed per IS 800 Cl. 7 (compression) and Cl. 6 (tension). OsdagBridge cross-bracing module used.

5.4 End Diaphragm Design

End diaphragms at the supports transfer transverse loads to the bearings, restrain the bottom flanges against lateral displacement, and maintain the girder cross-section geometry during construction and in service. They are designed per IS 800:2007 and IRC 24:2010 Cl. 507.

Table 5.26: End Diaphragm — Connection and Section Properties

Panel	Member	Connection	Section	A_g (mm²)	r_{min} (mm)
G1-G2	Diagonal	Welded	75x75x8		
G1-G2	Top / Bottom chord	Welded	65x65x6		
G2-G3	Diagonal	Welded	75x75x8		
G2-G3	Top / Bottom chord	Welded	65x65x6		

Note: A_g = gross cross-sectional area; r_{min} = minimum radius of gyration.

Table 5.27: End Diaphragm — Slenderness Ratio Check (IS 800 Cl. 3.8)

Panel	Member	Nature	Eff. Length KL (mm)	KL/r	Limit / Status
G1-G2	Diagonal	C	2920	142.0	250 — PASS
	Top chord	C	2500	118.0	250 — PASS
	Bottom chord	T	2500	118.0	400 — PASS
G2-G3	Diagonal	C	2920	142.0	250 — PASS
	Top chord	C	2500	118.0	250 — PASS
	Bottom chord	T	2500	118.0	400 — PASS

Note: IS 800 Cl. 3.8 — Limit $= 250$ *for compression members,* 400 *for tension members.* $K = 1.0$ *for members with both ends pinned.*

Table 5.28: End Diaphragm Design — Capacity Summary

Panel	Member	Section	Governing LC	Demand (kN)	Capacity (kN)	UR	Status
G1-G2	Diagonal	75x75x8	Basic ULS: 1.35DL + 1.5LL	185.000	340.000	0.540	PASS
	Chord	65x65x6	Basic ULS: 1.35DL + 1.5LL	95.000	210.000	0.450	PASS
G2-G3	Diagonal	75x75x8	Basic ULS: 1.35DL + 1.5LL	172.000	340.000	0.510	PASS
	Chord	65x65x6	Basic ULS: 1.35DL + 1.5LL	88.000	210.000	0.420	PASS

Note: Designed per IS 800 Cl. 7 (compression) and Cl. 6 (tension). OsdagBridge cross-bracing module used.

5.5 Overall Design Check Summary

Table 5.29: Overall Design Check Summary — All Members

Member / Check	Governing Load Combo	Demand	Capacity	UR
Girder — Moment	Basic ULS: 1.35DL + 1.5LL	2450.00 kNm	3000.00 kNm	0.82
Girder — Shear	—	420.00 kN	690.00 kN	0.61
Girder — LTB (constr.)	—	1620.00 kNm	3000.00 kNm	0.54
Girder — Deflection	Basic ULS: 1.35DL + 1.5LL	38.00 mm	75.00 mm	0.51
Girder — Stress	—	168.00 MPa	350.00 MPa	0.48
Girder — Fatigue	—	55.00 MPa	140.00 MPa	0.39
Transverse Shear (slab)	—	—	—	—
Crack Width (slab)	Frequent SLS	0.180 mm	0.300 mm	0.60
Deck — Flexure (sagging)	Basic ULS: 1.35DL + 1.5LL	42.50 kN-m/m	63.00 kN-m/m	0.67
Deck — Flexure (hogging)	Basic ULS: 1.35DL + 1.5LL	38.00 kN-m/m	58.00 kN-m/m	0.66
Deck — Cantilever Overhang	Basic ULS: 1.35DL + 1.5LL	18.50 kN-m/m	29.00 kN-m/m	0.64
Deck — Punching Shear	Basic ULS: 1.35DL + 1.5LL	0.85 MPa	1.25 MPa	0.68
Deck — One-Way Shear	Basic ULS: 1.35DL + 1.5LL	95.00 kN/m	140.00 kN/m	0.68
Cross Bracing — Compression	Basic ULS: 1.35DL + 1.5LL	183.60 kN	340.000 kN	0.54
Cross Bracing — Tension	Basic ULS: 1.35DL + 1.5LL	120.40 kN	280.000 kN	0.43
Cross Bracing — Slenderness	—	142.0	250	0.57
End Diaphragm — Moment	Basic ULS: 1.35DL + 1.5LL	183.60 kN	340.000 kN	0.54

Continued on next page

Table 5.29: Overall Design Check Summary — All Members (continued)

Member / Check	**Governing Load Combo**	**Demand**	**Capacity**	**UR**
End Diaphragm — Shear	Basic ULS: 1.35DL + 1.5LL	120.40 kN	280.000 kN	0.43

Note: UR = Demand / Capacity. All values ≤ 1.0 *indicate passing checks. The governing check for each component is highlighted in the individual design check sections above.*

Utilization Ratio Visualisation

Governing demand/capacity ratios for the primary element families are plotted below. Bars are green when UR ≤ 1.0 (pass) and red when UR > 1.0 (fail).

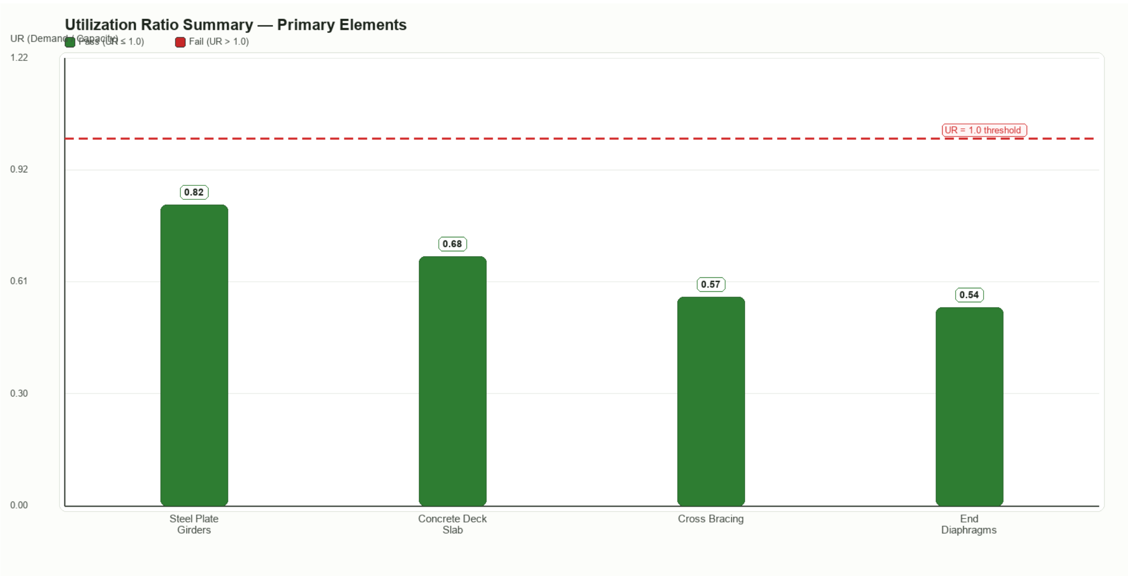

Figure — Utilization Ratio (Demand/Capacity) for Steel Plate Girders, Concrete Deck Slab, Cross Bracing and End Diaphragms. Red dashed line marks UR = 1.0 (pass/fail threshold).

6 Material Take-off & Quantity Summary

Table 7.1 Bill of Materials (Steel, Concrete, and Reinforcement Quantities)

Table 6.1: Bill of Materials (Steel, Concrete, and Reinforcement Quantities)

S.N.	Item Description	Volume	Quantity	Total Volume	Weight (MT)	Total Weight (MT)
1	Structural Steel (IS 2062) for Girders	V=A*L	4	1.620	3.150	12.600
2(a)	Cross Bracing - Top Chord	—	12	0.080	0.052	0.624
2(b)	Cross Bracing - Bottom Chord	—	12	0.075	0.049	0.588
2(c)	Cross Bracing - Diagonal Chord	—	24	0.120	0.039	0.936
3	Concrete (M40) for Deck Slab	L*B*t	1	48.500	121.250	121.250
4	Reinforcement Steel (Fe 500)	—	1	0.950	7.450	7.450
5	Shear Stud Connectors	—	800	0.020	0.001	0.800
6	Crash Barrier	—	2	3.200	4.000	8.000

Material Quantity Visualisation

The charts below summarise governing steel tonnage and concrete / reinforcement quantities from Table 7.1.

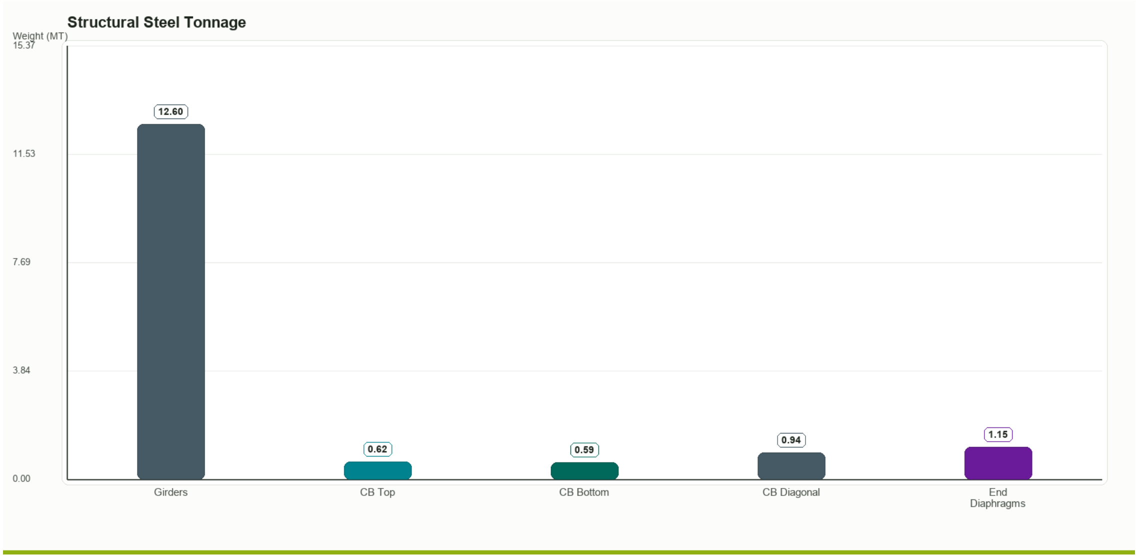

Figure — Structural steel tonnage (girders, cross bracing, end diaphragms).

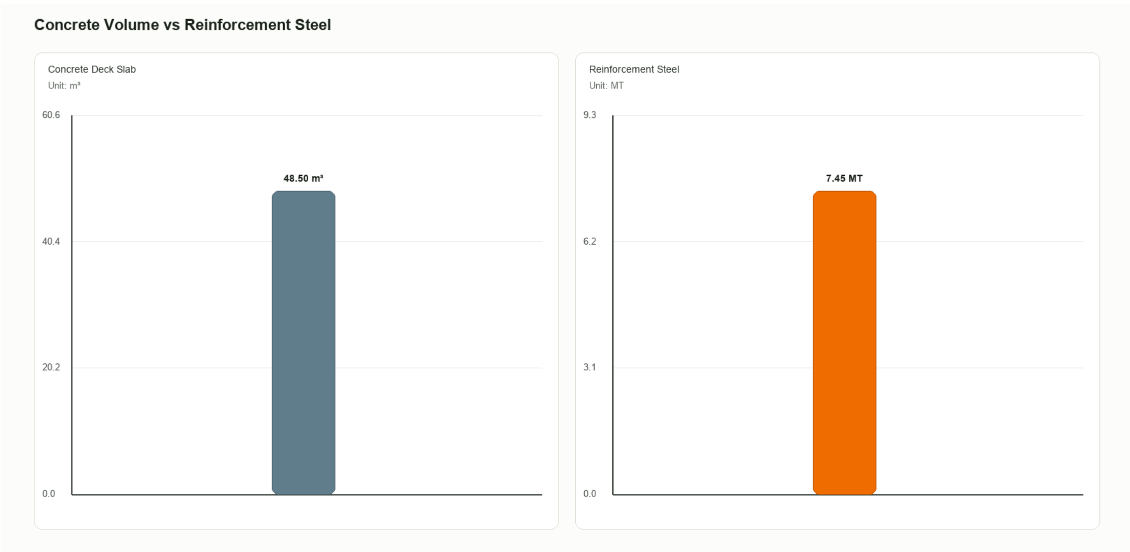

Figure — Concrete volume (m^3) and reinforcement steel (MT) shown on independent axes.

7 Standards & Assumptions

This section provides references to standards used to calibrate the OsdagBridge design modules, and notes the limitations of the current software version.

7.1 Design Standards

The following Indian Road Congress (IRC) codes and Indian Standards (IS) form the basis of all design calculations in this software.

Table 7.1: IRC Codes

Code	Year	Title / Scope
IRC 5	2015	General Features of Design - carriageway widths, kerb, footpath dimensions
IRC 6	2017	Loads and Load Combinations - dead load, live load, impact, wind, temperature, etc.
IRC 22	2015	Composite Construction (LS) - Composite section properties, ULS/SLS design, shear connectors
IRC 24	2010	Steel Road Bridges (LS) - Stiffener design, skew angle limits, diaphragm requirements
IRC 112	2020	Concrete Road Bridges - Deck slab flexure, shear, crack width, reinforcement
IRC SP 114	2018	Seismic Design of Road Bridges

Table 7.2: IS Codes

Code	Year	Scope
IS 800	2007	Steel construction - tension, compression, bending, shear, LTB, stiffeners, combined checks
IS 456	2000	Concrete - simplified stress-block for deck moment capacity
IS 1786	2008	Reinforcement steel properties
IS 1893 (Part 3)	2014	Earthquake resistant design
IS 2062	2011	Structural steel - yield and ultimate strength by grade

7.2 Analysis and Design Assumptions of This Version

Structural Analysis

- All girders are modelled as simply supported; continuous spans are not currently supported.
- A 3D grillage model (OSPGrillage) is used for load distribution. Grillage members carry composite section properties after the construction stage.
- The transverse member forces are computed using an approximate 2D frame analogy. For irregular or skewed geometries, a 3D FEM is recommended.
- Fixed bearing stiffness is modelled as $k = 1{,}000{,}000\,\text{kN/m}$ (virtually rigid); free/expansion bearing as $k = 100\,\text{kN/m}$.
- Construction stage sequence analysis is approximate. Detailed staged analysis should be performed for long-term deflection checks.

Composite Action

- Full shear connection is assumed at ULS with headed stud connectors designed per IRC 22:2015 Cl.606.
- Short-term composite section properties (modular ratio $n = E_s/E_{cm}$) are used for ULS checks.
- Long-term composite section properties (with creep-adjusted modular ratio) are used for SLS deflection and crack-width checks.
- Pre-composite stage: the steel girder alone resists all construction loads prior to concrete gaining strength.

Material Properties (IRC 22:2015 Annex III)

- Steel: $E_s = 200{,}000\,\text{MPa}$, $G_s = 80{,}000\,\text{MPa}$, $\nu = 0.30$, $\alpha = 11.7 \times 10^{-6}/°\text{C}$ (grade-independent).
- Minimum structural concrete grade: M25.
- Default reinforcement grade: Fe500.

Partial Safety Factors (IRC 22:2015 Cl.601.4)

- γ_{m0} (steel yield, ULS) = 1.10

- γ_{m1} (steel ultimate, ULS) = 1.25
- Reinforcement (ULS) = 1.15
- Welds – shop: 1.25; field: 1.50
- Fatigue (γ_{mft}) = 1.35

Loading

- Dead load densities per IRC 6:2017 Cl.203: structural steel 78.5 kN/m^3, concrete 25.0 kN/m^3, bituminous wearing course 24.0 kN/m^3.
- Multi-lane live load reduction factors per IRC 6:2017 Cl.204.4 Table 6A: 1st lane = 1.0, 2nd lane = 0.8, 3rd lane onwards = 0.4.
- Impact factor (dynamic load allowance) computed from span per IRC 6:2017 Cl.208.2/208.3.
- Wind load applied as transverse, longitudinal, and vertical components per IRC 6:2017 Cl.209.3.3–209.3.5.

Serviceability Limits

- Deflection limits (IRC 22:2015 Cl.604.3.2): live load + impact $\leq L/800$; total $\leq L/600$.
- SLS stress limits: concrete $\sigma_c \leq 0.48 f_{ck}$; rebar $\sigma_s \leq 0.80 f_{yk}$; steel $f_e \leq 0.9 f_y$.
- Permissible crack width: $w_k \leq 0.3$ mm (bridge deck, exposure class XS2/XD2 per IRC 112:2020 Cl.12.3.2).

Fatigue

- Reference fatigue life: $N_{sc} = 2 \times 10^6$ cycles (IRC 22:2015 Cl.605).
- Constant stress range is assumed. A thickness correction factor μ_r is applied for plate thickness > 25 mm.

Stiffener Design

- Intermediate transverse stiffener and bearing stiffener design follows IS 800:2007 Cl.8.7.2/8.7.3 and IRC 24:2010 Cl.509.7.2/509.7.3.

7.3 Known Limitations of This Version

- Substructure (piers, pile caps, foundations) and bearing design are not included.
- Splice connection design is not implemented.
- Skew angle > 15 degrees requires independent manual analysis (IRC 24 Cl. 504.8).
- Construction stage sequence analysis is approximate; detailed staged analysis should be performed for long-term deflection checks.
- The grillage analysis assumes simply supported boundary conditions; continuous spans are not currently supported.

References

1. IRC 5 (2024) — *Standard Specifications and Code of Practice for Road Bridges, Section I: General Features of Design.*
2. IRC 6 (2017) — *Standard Specifications and Code of Practice for Road Bridges, Section II: Loads and Load Combinations.*
3. IRC 22 (2014) — *Standard Specifications and Code of Practice for Road Bridges, Section VI: Composite Construction (Limit State Design).*
4. IRC 24 (2010) — *Standard Specifications and Code of Practice for Road Bridges, Section V: Steel Road Bridges (Limit State Method).*
5. IRC 112 (2011) — *Code of Practice for Concrete Road Bridges.*
6. IRC SP 114 (2018) — *Guidelines for Seismic Design of Road Bridges.*
7. IS 800 (2007) — *Indian Standard: General Construction in Steel — Code of Practice.*
8. IS 2062 (2011) — *Hot Rolled Medium and High Tensile Structural Steel — Specification.*
9. Subramanian, N. (2008). *Design of Steel Structures.* Oxford University Press.
10. Steel-INSDAG Teaching Resource Materials. `https://www.steel-insdag.org`